Optimizing Inventory and Sales Forecasting at Pastry palace: A Data-Driven Approach to be Efficient and Organised

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled "**Optimizing Inventory and Sales Forecasting at Pastry palace: A Data-Driven Approach to be Efficient and Organised**". I extend my appreciation to Pastry Palace for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

Signature of Candidate:

Name: Aryan Kumar

Date: 10-06-2025

1. Executive Summary

This project focuses on Pastry Palace, a small-scale food retail snacks and bakery shop located in Indira complex, Central Hospital Road, Dakra, Khalari, Churi, Jharkhand 829210, . The business has been running since August 2021 and is located in a semi-rural area, serving local walk-in customers. It offers a variety of freshly made items such as kachori, pizza, momos, handmade cakes, and biscuits, along with selected cold drinks and packaged FMCG products. While the business decisions rely on the owner's experience it currently lacks the tracking and managing of sales and stocks of the perishable and non perishable items.

This project will follow multiple steps to help the business understand the benefits of decision making through the data insights rather than random guess based on experience. Excel and python will be used to present the trends, insights , patterns in sales, product demand, wastage, and profitability. Techniques such as ABC analysis, profit margin comparison, and inventory movement tracking will be applied to generate meaningful reports.

The expected outcome is to reduce daily wastage, improve inventory management, and help the owner make better decisions that increase profitability and reduce stock-related losses.

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2. Organization Background

Pastry Palace is a small bakery and snacks shop established on **20 August 2021**. It is owned by **Mrs. Sunita Agarwal** and managed by her with the support of **three in-store staff**, **one bakery chef**, and **one helper**. The shop operates in the **Food Retail – Snacks, Bakery, and FMCG sector**, following a **B2C (Business-to-Consumer)** model that serves local walk-in customers in a semi-rural area.**The store operates daily from 8:30 AM to 9:45 PM**, catering to both morning snack buyers and evening footfall.

The business began with an initial investment of **₹5 lakh**, followed by an additional **₹1 lakh** added in small installments to support expansion and stability. After that, the business continued to run with internal revenue circulation, and no further external investment was made. In the first **six months**, the shop faced **consistent losses**, mainly due to **limited customer traffic and pricing challenges**. Although customer demand has improved over time, the business continues to face challenges in **tracking inventory, managing perishable items**, and **identifying high-profit products**.

This project aims to collect and summarize raw sales and stock data with the help of the owner in order to uncover meaningful insights. The goal is to reduce wastage, prevent stockouts, and improve overall profitability through data-driven decisions.

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3. Problem Statement

3.1 The business lacks a proper system to track sales and inventory, making it difficult to identify fast- and slow-moving items, manage perishables, and avoid stockouts.

3.2 There is no product-wise profit tracking, which limits the owner's ability to make data-driven pricing and production decisions, ultimately affecting profitability.

4. Background of the Problem

At **Pastry Palace**, decisions regarding how much to prepare, which items to restock, and which products to showcase or promote are made based on gut feeling, intuition, and experience rather than data or sales insights. Records are not documented in any formal manner and are instead managed verbally.

In discussions with the owner, several recurring challenges were identified: frequent wastage of unsold perishable items, missed profit due to high demand for low-margin products, stockouts of popular items, and lack of visibility into which products actually contribute to profit.

One example shared by the owner involves how she handles a high-demand, low-margin ice cream brand. Although she stocks multiple cartons, she sometimes delays opening the second pack—or tells customers the product is finished—so that they might choose other similar items with higher margins. This highlights how product visibility and sales decisions are currently driven more by experience and perceived margins rather than by actual data on stock levels, sales trends, or profitability.

Also, I personally noticed that the shop has a lot of product variety, especially in snacks and bakery items. It can feel a bit overwhelming to choose, and from a business point of view, this might lead to confusion for customers and make it harder to manage stock or focus on what's actually selling.

Introducing a system to record daily sales and stock movement, followed by analyzing trends and margins using tools like Excel, will help the owner make smarter decisions. It could lead to more profitable sales, reduced wastage, and better inventory control.

5. Problem Solving Approach

The Problems solution and and improvements that I am trying to achieve will be done in the following way-

5.1. Data Collection with Owner's Support - Raw sales data and inventory-related observations will be collected manually, since no formal records exist. The owner will assist in estimating daily product-wise quantity prepared, sold, and wasted.

5.3. Sample set of data fields - Collected data will be structured using Microsoft Excel. Fields such as Item Name, Date, Category, Quantity Prepared, Quantity Sold, Wastage, Cost Price, and Selling Price will be tabulated. Simple Excel formulas will be used to calculate daily profit/loss, waste percentage, and trends.

5.2. The following techniques will be implemented through Excel and Python:

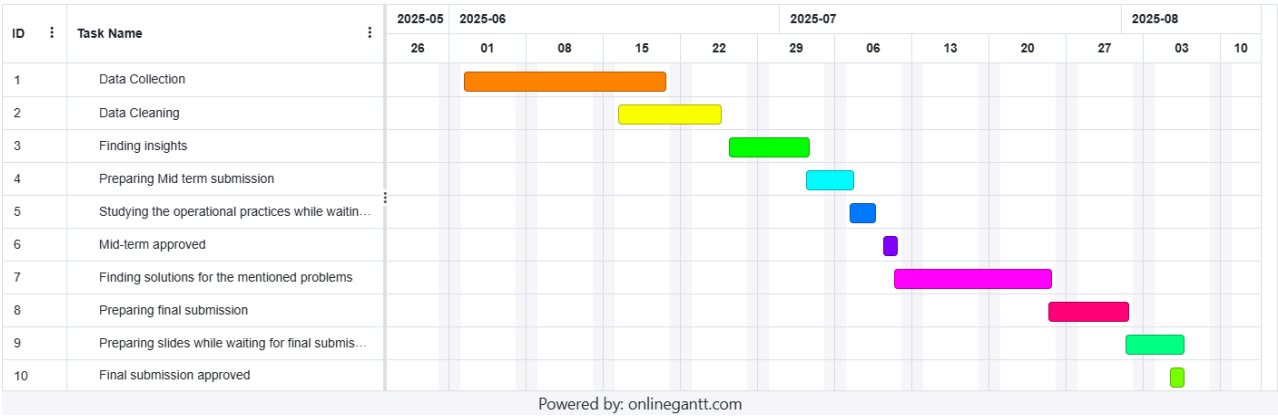
- ABC Analysis: To categorize products into high, medium, and low-value categories based on total revenue contribution, helping focus on best-sellers.
- Profit Margin Analysis: To calculate product-wise profit and identify high-margin vs. low-margin items.
- Wastage Tracking: To monitor daily leftover items and estimate losses.
- Stock Alert Logic: Using reorder point techniques, simple alerts will be built in Excel to highlight when fast-moving items are running low.

5.3. Practical Implementation - The owner will be shown how to update the Excel sheet daily. The system is designed to be easily maintainable without any technical expertise.

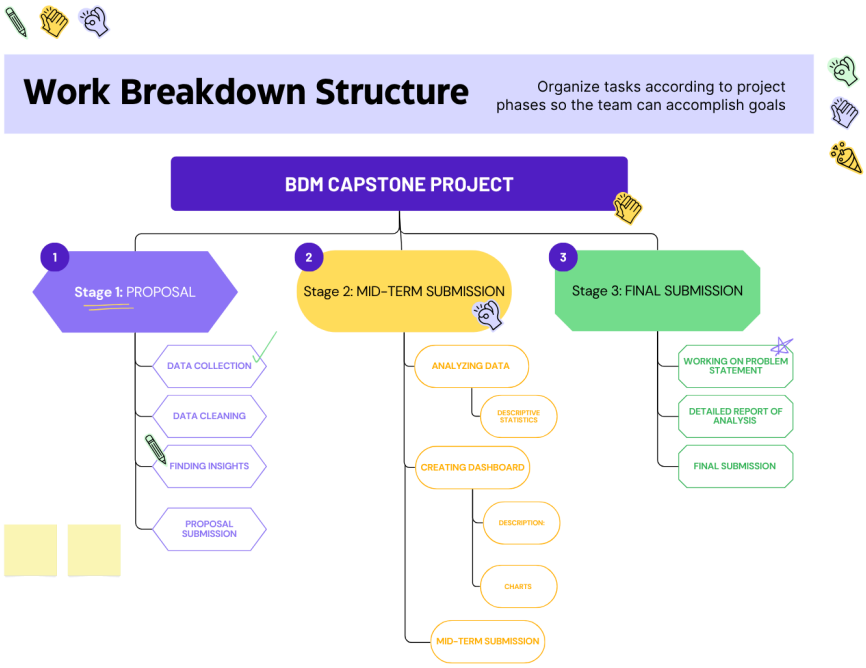
5.4. Expected Benefits - This approach is expected to reduce wastage, improve inventory rotation, and help the owner prioritize high-profit products. Over time, it will build confidence in data-driven decision-making and reduce dependence on guesswork.

6. Expected Timeline

GANTT CHART



WORK BREAKDOWN STRUCTURE



7. Expected Outcome

- Inventory planning will improve by tracking what gets sold and what gets wasted daily.

- The owner can focus more on high-margin items after knowing the profit of each product.

- A simple Excel-based system will help avoid guesswork and support better daily decisions.

- The project will make the business more organized and reduce dependence on memory or verbal records.

- Overall, it will help in saving money, increasing profits, and running the shop more smoothly.